National University of Computer and Emerging Sciences

**Lab Manual # 04**

**Digital Logic Design (EL1005)**

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| Section | BDS-2A |
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**OBJECTIVE:**

- To learn K-map and its usage in order to obtain cost effective circuit for implementation

**APPARATUS:** Logic trainer, Logic probe

**COMPONENTS:** ICs 74LS08, 74LS32, 74LS04

**THEORY:**

### Karnaugh Map

The Karnaugh map (K-map) is a method used to simplify Boolean expressions. K-Map is a grid- like representation of a truth table that gives more insight. The required Boolean results are transferred from a truth table onto a two-dimensional grid where the cells are ordered in gray code and each cell position represents one combination of input conditions, while each cell value represents the corresponding output value. Optimal groups of 1s or 0s are identified, which represent the terms of a canonical form of the logic in the original truth table. These terms can be used to write a minimal Boolean expression representing the required logic.

Karnaugh map is used to obtain optimized logic representation so that it can be implemented using a minimum number of logic gates. The sum-of-product form can always be implemented using AND gates feeding into an OR gate, and a product-of-sum form leads to OR gates feeding an AND gate.

The minimization will result in reduction of the number of gates (resulting from less number of terms) and the number of inputs per gate (resulting from less number of variables per term). The minimization will reduce cost and power consumption of the logic circuit.

**Question 1:** In a storage room, entrance via door is highly secured. Door can be opened by the following signals:

- Control Signal '**C**' gives '1' when Control Room allows.
- Pressure sensor '**P**' installed in front of doors, gives '1' when a person stands on it (pressing the sensor).
- Finger Print sensor '**F**' gives '0' when a fingerprint is successfully recognized.

The Door opens only under the following conditions:

- Door '**D**' is opened when Pressure sensor is pressed and fingerprint is successfully recognized.
- Door '**D**' can also be opened when control signal is received and pressure sensor is pressed.

**Your task is to fill in the truth table given below for the door:**

**NOTE: To open the door assign '1' in the following table.**

| **Inputs** | | | **Output** |
|---|---|---|---|
| **C** | **P** | **F** | **D** |
| 0 | 0 | 0 | |
| 0 | 0 | 1 | |
| 0 | 1 | 0 | |
| 0 | 1 | 1 | |
| 1 | 0 | 0 | |
| 1 | 0 | 1 | |
| 1 | 1 | 0 | |
| 1 | 1 | 1 | |

a. Fill the truth table for the output D
b. Write Boolean function for D in canonical form using minterms.
c. Write Boolean function for D in canonical form using maxterms
d. Find minimal SOP expression for D using K-map, draw circuit and implement on the Logic Trainer
e. Find minimal POS expression for D using K-map, draw circuit and implement on the Logic Trainer

**Question 2: For the Boolean function $F1(A,B,C,D)=\Sigma m(2,4,12,14)$ do the following:**

a. Draw Truth Table
b. Find minimal SOP expression for Boolean function $F1$ using K-map, draw circuit and implement on the Logic Trainer

**Question 3: For the Boolean function $F2(A,B,C,D)=\Pi M(9,13,15)$ do the following:**

a. Draw Truth Table
b. Find minimal SOP expression for Boolean function $F2$ using K-map, draw circuit and implement on the Logic Works
c. Find minimal POS expression for Boolean function $F2$ using K-map, draw circuit and implement on Logic Trainer

**Question 4: A Boolean functions is given as follows:**

**$F(A,B,C,D)=\sum m(2,4,6,11,14)$**
**$d(A,B,C,D)=\sum m(0,3,10,12)$**

Use **KMaps** to optimize the function into**:**

a. Sum of Products (SOP) Form, draw circuit and implement on the Logic Works
b. Product of Sums (POS) Form, draw circuit and implement on the Logic Works